

Movie
- id: int - title: String - duration: int
+ Movie(id: int, title: String, duration: int) + getId(): int + getTitle(): String + getDuration(): int + toString(): String

Theater
- id: int - name: String - totalRows: int - totalColumns: int
+ Theater(id: int, name: String, totalRows: int, totalColumns: int) + getId(): int + getName(): String + setName(name: String): void + getTotalRows(): int + setTotalRows(totalRows: int): void + getTotalColumns(): int + setTotalColumns(totalColumns: int): void + toString(): String

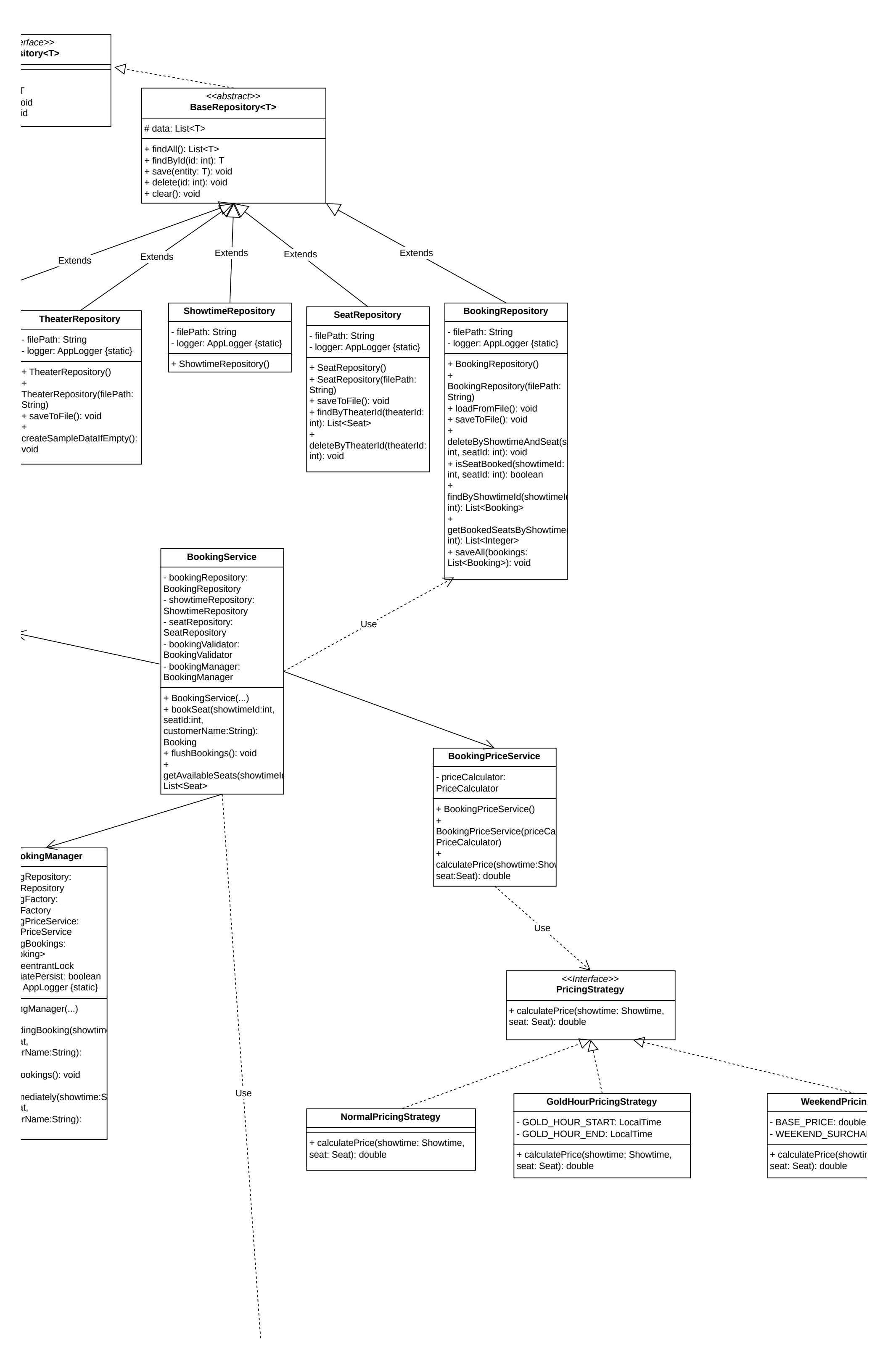
Seat
- id: int - theaterId: int - row: int - column: int

MovieRepository
- filePath: String - logger: AppLogger {static}
+ MovieRepository() + MovieRepository(filePath: String) + saveToFile(): void + createSampleDataIfEmpty(): void + findByName(title: String): Movie

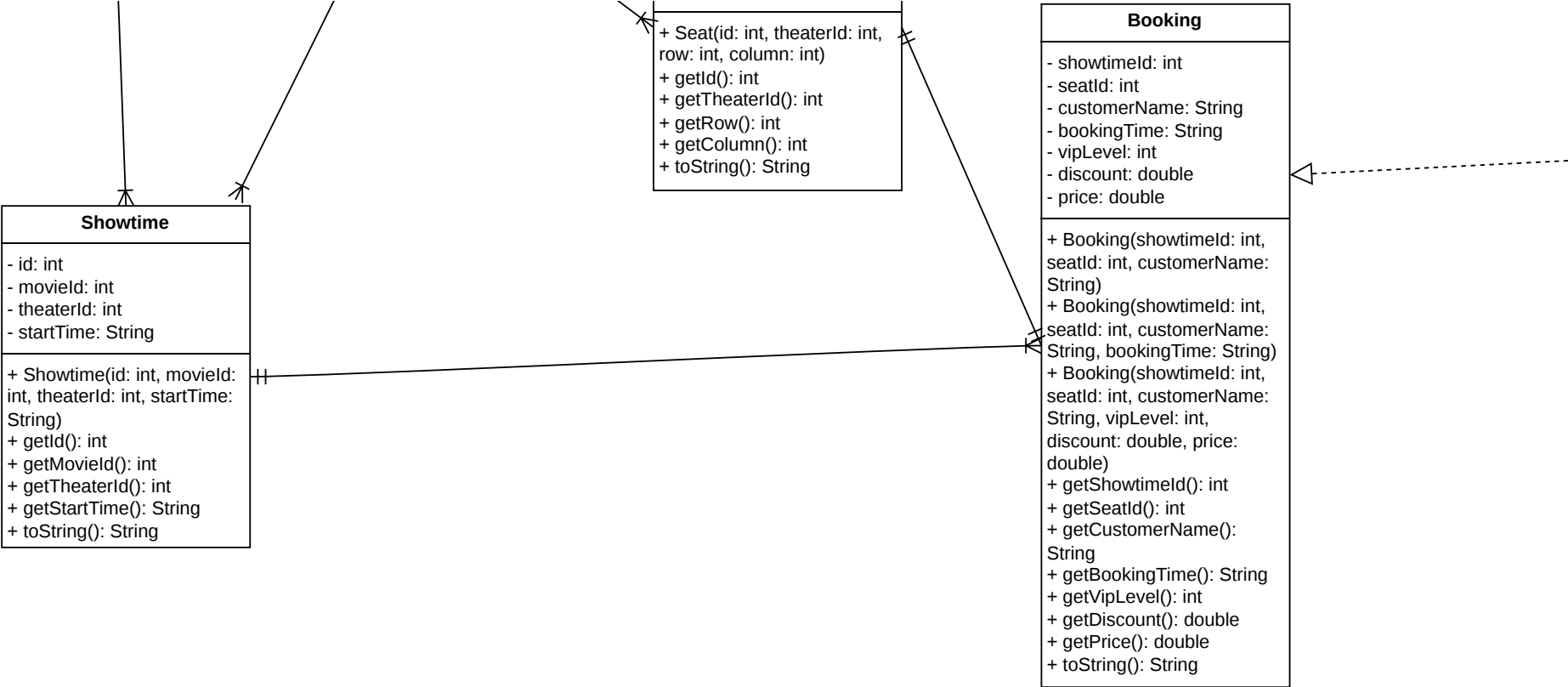
BookingValidator
- showtimeRepository: ShowtimeRepository - seatRepository: SeatRepository
+ BookingValidator(...) + validateBooking(showtimeId: int, seatId: int, customerName: String): Showtime

Booking
- bookingId: int - showtimeId: int - seatId: int - pending: boolean - lock: ReentrantLock - immediate: boolean - logger: AppLogger
+ BookingId() + addPendingSeat(seat: Seat, customerName: String): Booking + flushBooking()

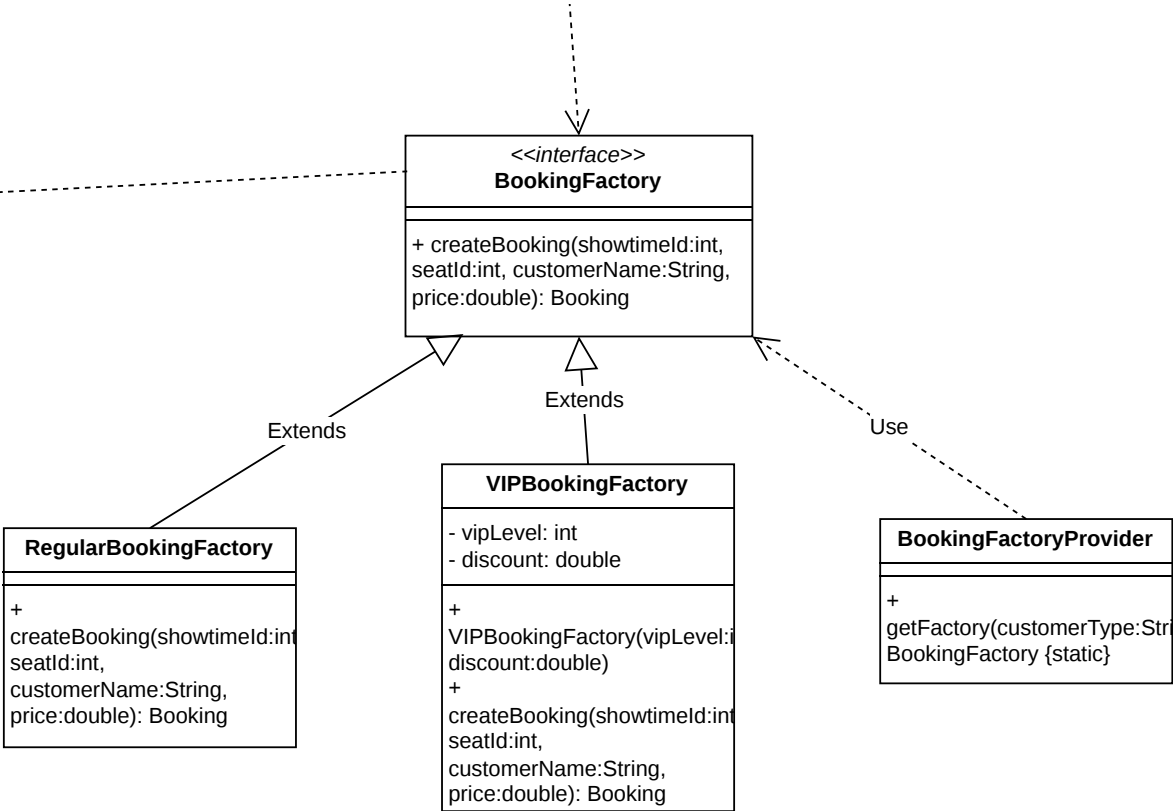
<<interface>> IRepository
+ findAll(): List<T> + findById(id: int): T + save(entity: T): void + delete(id: int): void



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FileStorage
- INSTANCE: FileStorage
+ getInstance(): FileStorage + readLines(filePath: String): List<String> + writeLines(filePath: String, lines: List<String>): void



Validator
+ validateCustomerName(name: String): void + validateSeatId(seatId: int, maxSeats: int): void + validateShowtime(showtimeId: int, repo: ShowtimeRepository): void + validateSeatInTheater(seatId: int, theaterId: int, repo: SeatRepository): void

AppConstants
+ DEFAULT_MAX_SEATS: int + MOVIES_FILE: String + BOOKINGS_FILE: String + THEATERS_FILE: String + SHOWTIMES_FILE: String + SEATS_FILE: String + BASE_TICKET_PRICE: double + GOLD_HOUR_PRICE: double + WEEKEND_SURCHARGE: double

AppLogger
- INSTANCE: AppLogger - logger: Logger
+ getInstance(): AppLogger + info(message: String): void + warning(message: String): void + severe(message: String): void + debug(message: String): void

AppContext
- INSTANCE: AppContext - movieRepository: MovieRepository - theaterRepository: TheaterRepository - seatRepository: SeatRepository - showtimeRepository: ShowtimeRepository - bookingRepository: BookingRepository - bookingFactory: BookingFactory - bookingValidator: BookingValidator - bookingPriceService: BookingPriceService - bookingManager: BookingManager - bookingService: BookingService
+ getInstance(): AppContext + getBookingService(): BookingService

